What is this?

This notebook is part of a collection of `pluto` notebooks on various topics discussed during the Time Domain Astrophysics course delivered by Stefano Covino at the Università dell'Insubria in Como (Italy). Please direct questions and suggestions to stefano.covino@inaf.it.

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Time of Arrival analysis

- So far, we have assumed that the our data are constituted by a set of N data points $(t_j, y_j), j = 1, \ldots, N$, possibly with known uncertainties.
- We can anyway think to different datasets. For instance, at X-ray and shorter wavelengths, individual photons are detected and background contamination is often negligible.
    - In such cases, the data set consists of the arrival times of individual photons $t_j, j = 1, \ldots, N$, or, in principle and more generally, of the time of occurrence of a given phenomenon.
- Given such a data set, how do we search for a periodic signal, and more generally, how do we test for any type of variability?

Tip

- Even well beyond astronomy, dealing with these kind of datasets presents problems of great interest.

Rayleigh Test

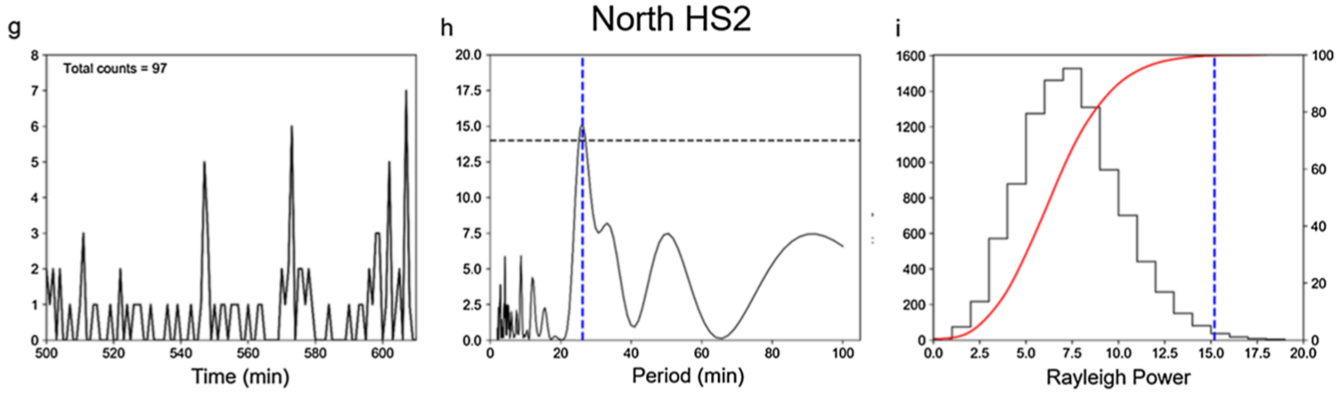
- The best known classical test for variability in arrival time data is the *Rayleigh test*.
- Given a trial period, P, the phase, φ_j, corresponding to each datum is evaluated using:

$$\varphi = \frac{t}{P} - \mathrm{int}\left(\frac{t}{P}\right)$$

- And the following statistics is computed:

$$R^2 = \left(\sum_{j=1}^{N} \cos(2\pi\varphi_j)\right)^2 + \left(\sum_{j=1}^{N} \sin(2\pi\varphi_j)\right)^2$$

- A simple way to read this expression is referring to a random walk.
    - Each angle φ_j defines a unit vector, and R is the length of the resulting vector.
- For random data R^2 is small, and for periodic data R^2 is large when the correct period is chosen.
- R^2 is evaluated for a grid of periods, and the best period is chosen as the value that maximizes R^2.
- For $N > 10$, $2R^2/N$ is distributed as χ^2 with two degrees of freedom. This easily follows from the random walk interpretation.
- Then, one can assess the significance of the best-fit period as the probability that “a value that large” would happen by chance when the signal is stationary.



- This plot (from Weigt et al. 2019) shows a time-series of event times, the Rayleigh periodogram, and simulations to infer the significance of the periodogram maximum.
- It is also possible to generalize the Rayleigh test including multiple harmonics in the time of arrival decomposition:

$$R_n^2 = \frac{2}{N}\sum_{k=1}^{n}\left[\left(\sum_{j=1}^{N}\cos(k\varphi_j)\right)^2 + \left(\sum_{j=1}^{N}\sin(k\varphi_j)\right)^2\right]$$

- n is of course the number of harmonics.

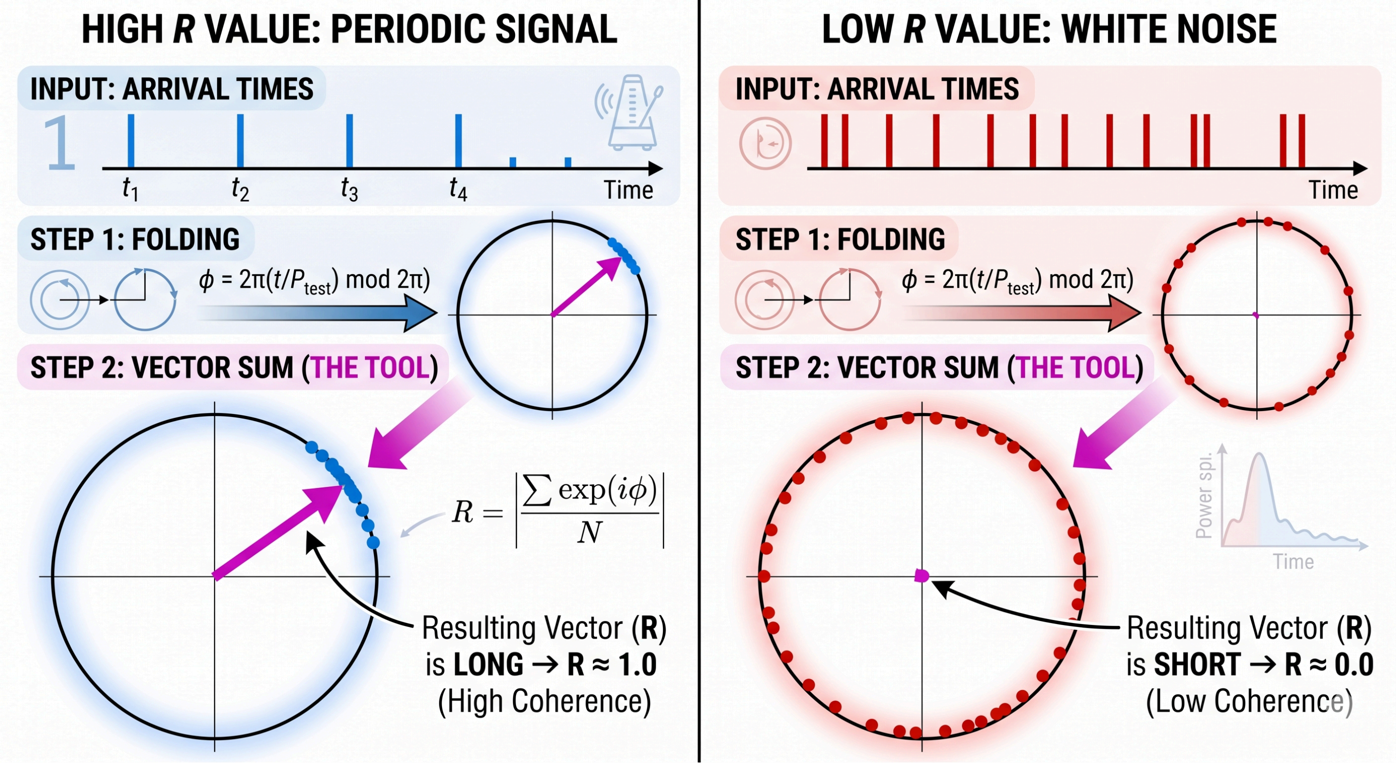
Tip

To help the understanding

- To understand the Rayleigh periodogram, it's best to think of it as a test for "circular uniformity."
- When you have a set of arrival times, you wrap them around a circle based on a test period P.The Rayleigh statistic (R) is the length of the average vector of all those points on the circle.

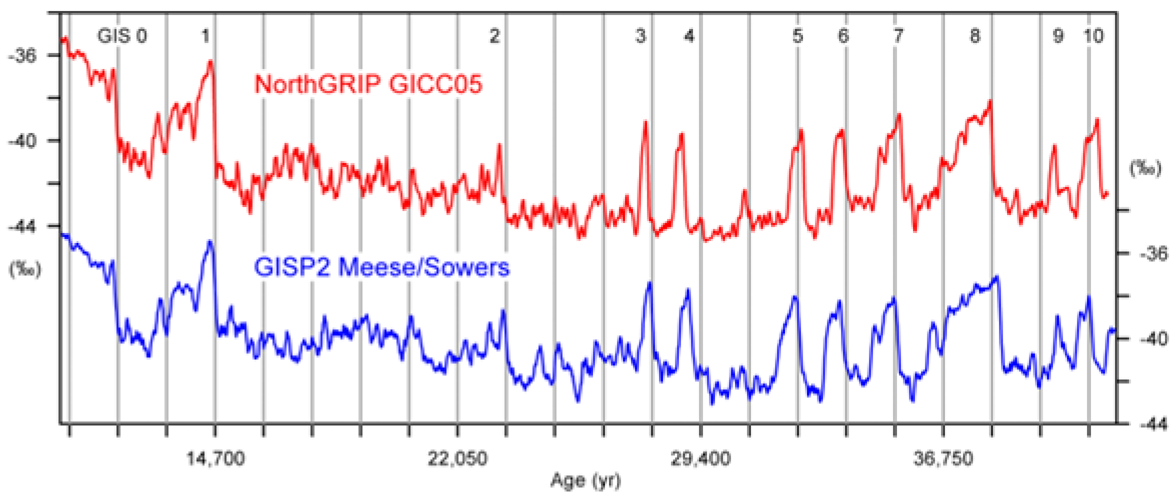
- High R values, even approaching 1.0, can occur when the arrival times are strongly periodic. If your test period matches the actual cycle of the data, the events will all "clump" together at the same phase on the circle.
  - Imagine a pulsar emitting a beam every 1.0 seconds. Your arrival times are: 1.02s, 2.01s, 3.03s, 4.02s, 5.01s, etc.
  - If you test a period of $P = 1.0s$, every single data point falls at roughly the "12 o'clock" position on your phase circle. Because they are all pointing in the same direction, their vectors add up constructively.

- Low R values occur when the arrival times are stochastic (random) and/or if your test period P is completely wrong for the data.
  - Imagine rain falling on a tin roof. The hits are random and independent: 0.42s, 1.15s, 2.89s, 3.21s, 4.77s, etc.
  - When you wrap these random times around a circle, the phases are scattered everywhere and when you add these vectors together, they cancel each other out.



Exercize: DO-climate- events

- In a paper by Ditlevsen et al. 2006, a possible periodicity of about 1500 years for the Dansgaard-Oeschger (DO) events, observed in the Greenland ice cores is discussed.
- Dansgaard–Oeschger events are rapid, dramatic climate fluctuations that occurred repeatedly during the last glacial period (roughly 115,000 to 11,000 years ago). They are essentially "flickers" in the Earth's climate system where temperatures in the North Atlantic region spiked and then crashed back to glacial conditions.
- The warming phase was incredibly fast, often jumping by 8°C to 16°C in Greenland within just a few decades or sometimes even less than a decade.
- There were several of these events during the last ice age. While they seem to recur roughly every 1,470 years, specialists still debate whether they are strictly periodic or more random.
- The leading theory involves a massive "reset" of the Atlantic Meridional Overturning Circulation (AMOC). the giant conveyor belt of ocean currents that brings warm water to the North Atlantic.



- The $\delta^{18}O$ isotope records from NGRIP and GISP on their stratigraphic time scale. The vertical bars are separated by 1470 years. The analysis focus on the well defined fast onsets of DO events, which are transitions from the stadial to the interstadial states. Ages are b2k=BP+50 years.
- B.P. (Before the Present) is the number of years before the present. Because the present changes every year, archaeologists, by convention, use A.D. 1950 as their reference. So, 2000 B.P. is the equivalent of 50 B.C.
- In geochemistry, paleoclimatology and paleoceanography $\delta^{18}O$, or delta-O-18, is a measure of the ratio of stable isotopes oxygen-18 (^{18}O) and oxygen-16 (^{16}O). It is commonly used as a measure of the temperature of precipitation, as a measure of groundwater/mineral interactions, and as an indicator of processes that show isotopic fractionation, like methanogenesis. In paleosciences, ^{18}O:^{16}O data from corals, foraminifera and ice cores are used as a proxy for temperature.

- Identifiying a DO event is not an easy task, and there are discussions about the actual definition of what a "rapid fluctuation" is. Of course, we do not enter in this discussion, and adopt the list reported by Rahmstorf (2003).

```
DOevts = [11605, 13073, 14630, 23398, 27821, 29021, 32293, 33581, 35270, 38387, 41143,
42537, 45362];
```

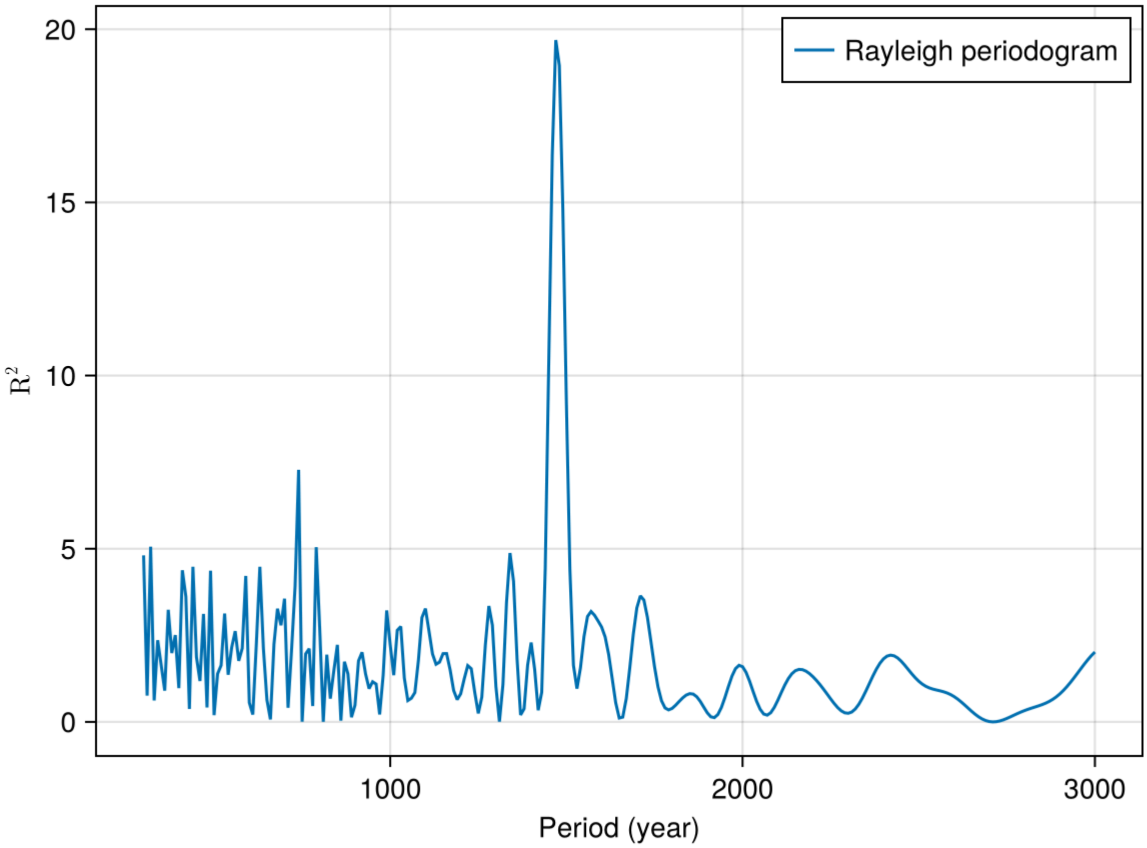
- Let's now code a function to compute the Rayleigh periodogram:

```
z2n (generic function with 1 method)
```

```
function z2n(freqs, time; harm=1)
    N = length(time)
    Z2n = []
    for ni in freqs
        aux = 0
        for k in 1:harm
            Phi = mod.(ni .* time,1)
            arg = k .* Phi*2.0*π
            phicos = cos.(arg)
            phisin = sin.(arg)
            aux = aux .+ (sum(phicos)^2 + sum(phisin)^2)
        end
        push!(Z2n,(2.0/N)*aux)
    end
    return Z2n
end
```

- We study possible periods from 300 to 3000 years.

```
begin
    per = range(start=300,stop=3000,step=10)
    freq = 1 ./ per
    res = z2n(freq,DOevts)
end;
```



- Given that with large N (and indeed this is NOT the case) the Rayleigh statistics follow the χ^2 distribution with two degrees of freedom, it is easy to commpute the FAP for the observed periodogram peak.
- For large N, given the definition of the Rayleigh periodogram, FAP $\sim e^{-Z}$, where Z is the power. Please, be aware that for a very low number of points we are likely over-estimating the FAP.
- Evaluating the number of independent periods (or frequencies) allowed by the input data is more difficult. In order to keep everything we might simply assume $N_{eff} \sim N_{period}/2$.

```
begin
    pmax,pidx = findmax(res)
    lfap = exp(-pmax/2)
    gfap = 1-(1-lfap)^(length(per)/2.)
end;
```

In our case, the periodogram maximum is 19.7 at period 1470.0 years

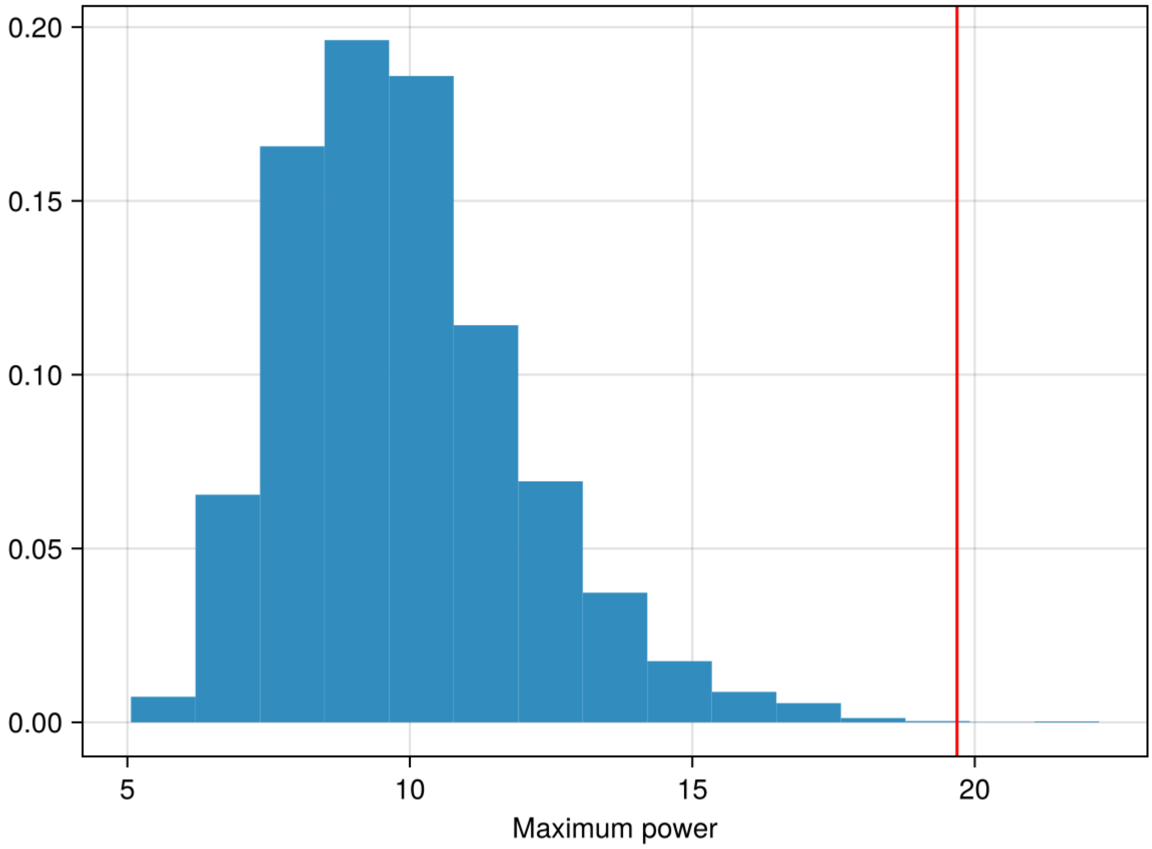
Local FAP 0.005% and global FAP 0.7%

- Thus, we have a maximum for a period $P \sim 1500$ years, its local significance (given the assumptions discussed above) is ∼99.99%, while the global significance is ∼ 99.28%

> Interesting, but crucially dependent on the number of independent frequencies that for our small dataset is difficult to evaluate and affected by the low number of events that makes the statistics probably unreliable.

- Let's check the results by a simple MonteCarlo approach, generating a large set of uniformly distributed time of arrivals

```
begin
    resb(x) = maximum(z2n(freq,x))
    bres = []
    for i in 1:10000
        artdt = rand(minimum(DOevts):maximum(DOevts),length(DOevts))
        push!(bres,resb(artdt))
    end
end
```



- And the observed value has a FAP (under the quite simple null hypothesis we have adopted) of 0.04%.
- Indeed, a value not too far from what (grossly) derived analytically.
- It is thus clear that the obained power is of interest, deserving further investigations.

Warning:

This is an interesting exercize, yet be aware that a MonteCarlo or bootstrap analysis for a time-series is a complex issue.

- Now, in order to go on, it is useful to recap some of the main features of the Poisson distribution

Poisson Distribution: a brief recap

- A Poisson Process is a model for a series of discrete event where the average time between events is known, but the exact timing of events is random. The arrival of an event is independent of the event before (waiting time between events is memoryless).
- A Poisson Process meets the following criteria (in reality many phenomena modeled as Poisson processes don't meet these exactly):
  - Events are independent of each other. The occurrence of one event does not affect the probability another event will occur.
  - The average rate (events per time period) is constant.
  - Two events cannot occur at the same time.
- The last point means an event either happens or not!

- A common examples of Poisson processes are customers calling a help center, visitors to a website, radioactive decay in atoms, photons arriving at a space telescope, and movements in a stock price.
- Poisson processes are generally associated with time, but they do not have to be. In the stock case, we might know the average movements per day (events per time), but we could also have a Poisson process for the number of trees in an acre (events per area).
- One instance frequently given for a Poisson Process is bus arrivals (or trains or Ubers). However, this is not (always) a true Poisson process because the arrivals are not independent of one another. Even for bus systems that do not run on time, whether or not one bus is late affects the arrival time of the next bus.

- The Poisson Distribution probability mass function gives the probability of observing k events in a time period given the length of the period and the average events per time:

$$P((\text{k events in a time period}) = e^{-\frac{\text{event}}{\text{time}} \times \text{ time period}} \frac{\left(\frac{\text{event}}{\text{time}} \times \text{ time period}\right)^k}{k!}$$

- $\textbf{events / time} \times \textbf{time period}$ is usually simplified into a single parameter, λ, the rate parameter:

$$P(\text{k events in interval}) = e^{-\lambda} \times \frac{\lambda^k}{k!}$$

- λ can be thought of as the *expected number of events in the interval*.

Waiting time

- The probability of waiting a given amount of time between successive events decreases exponentially as the time increases. The following equation shows the probability of waiting more than a specified time:

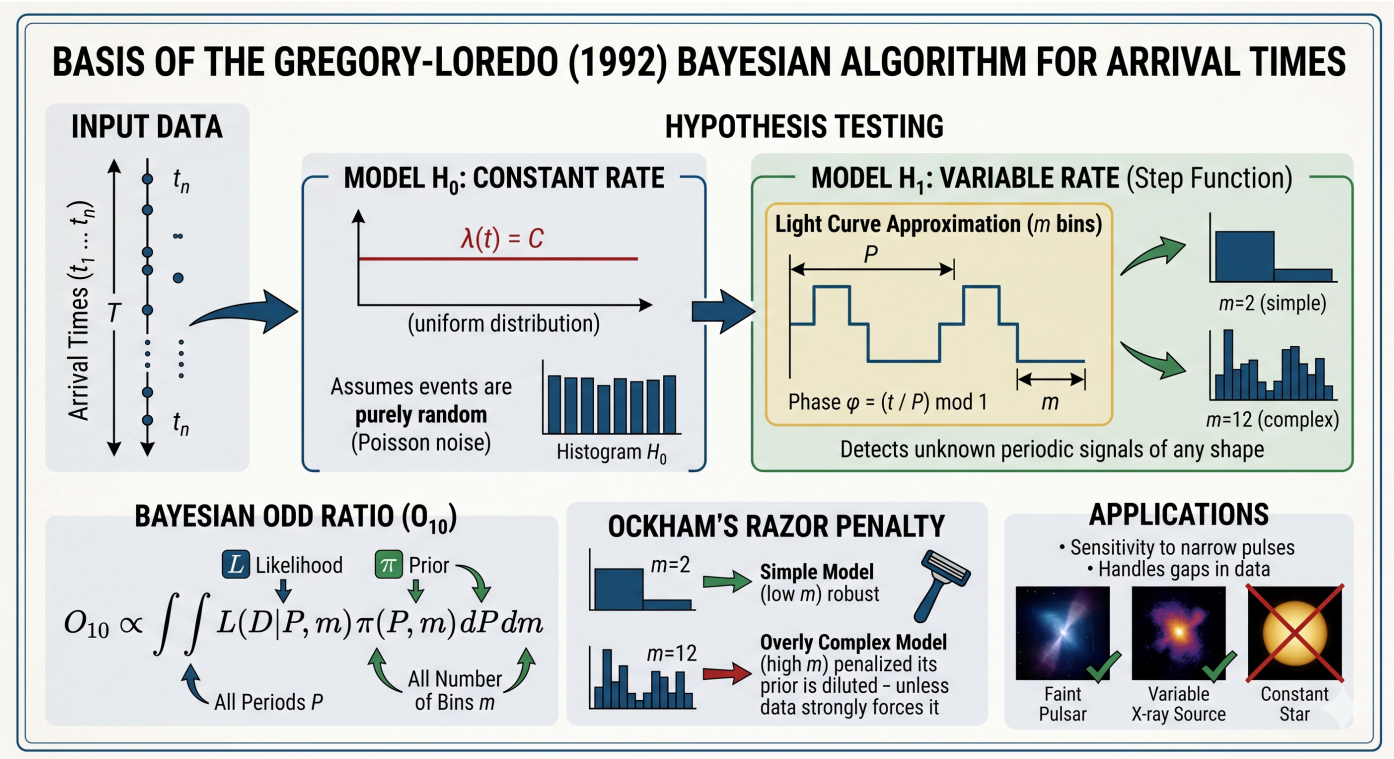
$$P(T > t) = e^{-\frac{\text{event}}{\text{time}} \times t}$$

- Conversely, the probability of waiting less than or equal to a time:

$$P(T \leq t) = 1 - e^{-\frac{\text{event}}{\text{time}} \times t}$$

The Gregory & Loredo algorithm

- In 1992 Gregory & Loredo pusblished a seminal paper describing a fully Bayesian procedure to analyse "time of arrival" data. We give here only a short description of the algorithm.
- The core of the algorithm is a comparison of two hypotheses:
  1. H_0 (The Null Hypothesis): The event arrival rate is constant (uniform distribution).
  2. H_1 (The Periodic/Variable Hypothesis): The event rate follows a "stepwise" distribution with m bins per period.
- The algorithm divides a trial period into m bins. It then varies m (typically from 2 to some m_{max}) to allow for different levels of signal complexity.
- For each number of bins (m), it calculates the likelihood of the observed distribution of events ($n_1, n_2, \ldots n_m$) and then uses Bayes' Theorem to calculate an "odds ratio", the probability that the source is variable versus the probability that it is constant.



- Let's divide the time interval covering our observations, $T = t_N - t_1$, into many arbitrarily small steps, Δt, so that each interval contains either 1 or 0 detections.
- Given an event rate, $r(t)$, the expectation value for the number of events during Δt is:

$$\mu(t) = r(t)\Delta t$$

- Poisson statistics tells us that the probability of detecting no events during Δt is:

$$p(0) = e^{r(t)\Delta t}$$

- Analogously, the probability of detecting a single event is:

$$p(1) = r(t)\Delta t \; e^{r(t)\Delta t}$$

- We can now compute the total likelihood, i.e. computing the probability of having observed a given sequence of events as:

$$p(D|r, I) = (\Delta t)^N e^{-\int_{(T)} r(t)dt} \prod_{j=1}^{N} r(t_j)$$

- For simplicity, we assume that the data were collected in a single stretch of time with no gaps. But it is possible to generalize the algorithm.

- With an appropriate $r(t)$, and priors for the model parameters, analysis of arrival time data is no different than any other model selection and parameter estimation problem.
- Instead of fitting a parametrized model, such as a Fourier series, here a nonparametric description of the rate function $r(t)$.
- The model includes the frequency ω (or period), a phase offset, the average rate A, and $M-1$ parameters for the rate.
- Marginalizing the resulting pdf one can produce
    1. an analog of the periodogram,
    2. expressions for computing the model odds ratio for signal detection, and
    3. for estimating the light curve shape.
- In the case when little is known about the signal shape, this method is superior to the more popular Fourier series expansion.
- A fairly detailed lecture about the Gregory & Loredo (1992) algorithm can be found here (notebook, html).

Reference & Material

Material and papers related to the topics discussed in this lecture.

- Ivezić et al. (2020) - "Statistics, Data Mining, and Machine Learning in Astronomy"
- Gregory & Loredo (1992) - "A New Method for the Detection of a Periodic Signal of Unknown Shape and Period"

Further Material

Papers for examining more closely some of the discussed topics.

- Ditlevsen et al. (2006) - "The DO-climate events are noise induced: Statistical questioning of the 1470 years cycle"
- Weigt al. (2019) - "Observations of Jupiter's auroral emission during Juno apojove 2017".
- Bao & Li (2020) - "Periodic X-ray sources in the Galactic bulge: application of the Gregory-Loredo algorithm"
- Bao & Li (2021) - "Searching for quasi-periodic oscillations in active galactic nuclei of the Chandra Deep Field South"

Course Flow

	Previous lecture	Next lecture	Course Summary
notebook	Science case about AGN and Blazars	Science case about FRBs	Course Summary
html	Science case about AGN and Blazars	Science case about FRBs	Course Summary

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